

Experiment 06

Application of K-Means and Hierarchical Clustering for Customer Segmentation

Dataset: Mall Customer Segmentation Dataset

Dataset Link:

<https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python>

Dataset Description

The Mall Customer Segmentation dataset contains demographic and behavioral information of customers visiting a shopping mall. The dataset is commonly used to analyze customer purchasing patterns and group similar customers into clusters using unsupervised machine learning techniques.

The dataset consists of **200 records** and **5 attributes**.

Features in the dataset:

CustomerID – Unique identifier assigned to each customer

Gender – Gender of the customer (Male or Female)

Age – Age of the customer in years

Annual Income (k\$) – Annual income of the customer measured in thousands of dollars

Spending Score (1–100) – A score assigned by the mall based on customer purchasing behavior

For clustering analysis, the following features were used:

Annual Income (k\$)

Spending Score (1–100)

These variables help identify spending behavior and economic capacity of customers, which are useful for segmentation analysis.

Mathematical Formulation of the Algorithm

K-Means Clustering

K-Means is an unsupervised learning algorithm that partitions the dataset into K clusters such that the total distance between data points and their respective cluster centroids is minimized.

Objective Function

$$J = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

Where

x_i = data point

μ_j = centroid of cluster j

K = number of clusters

Distance Metric (Euclidean Distance)

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Hierarchical Clustering

Hierarchical clustering builds a hierarchy of clusters using an agglomerative approach.

Steps:

1. Each data point initially forms its own cluster
2. The two closest clusters are merged together
3. This process continues until all data points are merged into one cluster

Ward linkage is used to minimize the increase in within-cluster variance when clusters are merged.

Algorithm Limitations

K-Means Limitations

- The number of clusters must be defined beforehand
- Sensitive to initialization of cluster centroids
- Sensitive to outliers and noise
- Works best for spherical clusters

Hierarchical Clustering Limitations

- Computationally expensive for large datasets
- Once clusters are merged they cannot be separated
- Sensitive to choice of linkage method
- Not suitable for very high-dimensional datasets

Methodology / Workflow

Step 1

Import required Python libraries including NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn.

```
[1]
✓ 2s
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.cluster import AgglomerativeClustering
from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import dendrogram, linkage
```

Step 2

Load the dataset Mall_Customers.csv using Pandas.

```
[2]
✓ 0s
df = pd.read_csv("Mall_Customers.csv")

df.head()
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Step 3

Inspect the dataset using functions such as:

```
df.head()
df.info()
df.describe()
```

```
[3] df.info()
df.describe()
```

*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
Column Non-Null Count Dtype
--- ---
0 CustomerID 200 non-null int64
1 Gender 200 non-null object
2 Age 200 non-null int64
3 Annual Income (k\$) 200 non-null int64
4 Spending Score (1-100) 200 non-null int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

Step 4

Select relevant features for clustering:

Annual Income (k\$)

Spending Score (1–100)

```
[4] X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

Step 5

Normalize the selected features using StandardScaler to ensure uniform scaling.

```
[5] scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

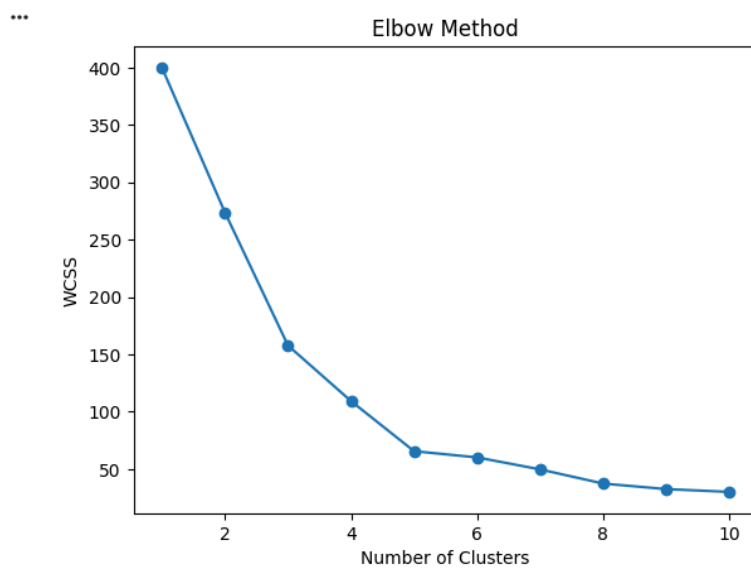
Step 6

Use the Elbow Method to determine the optimal number of clusters by plotting WCSS against different cluster values.

```
[6]
✓ Os
wcss = []

for i in range(1,11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1,11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```



Step 7

Apply the K-Means clustering algorithm using the optimal number of clusters.

```
[7]
✓ Os
▶ kmeans = KMeans(n_clusters=5, random_state=42)
  y_kmeans = kmeans.fit_predict(X_scaled)

  df['cluster'] = y_kmeans
```

Step 8

Visualize the clusters using scatter plots.

[8]
✓ 0s

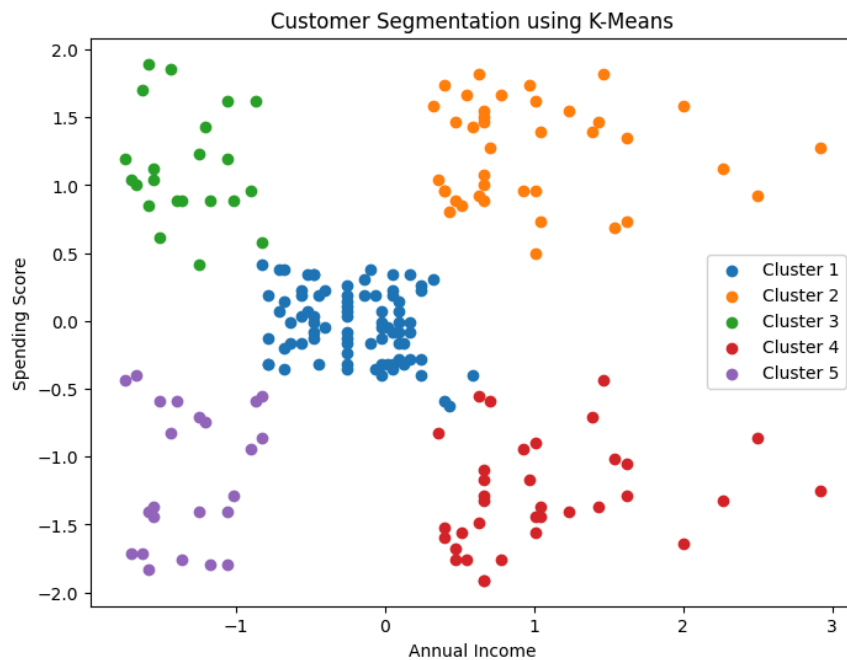
```
plt.figure(figsize=(8,6))

plt.scatter(X_scaled[y_kmeans==0,0], X_scaled[y_kmeans==0,1], label="Cluster 1")
plt.scatter(X_scaled[y_kmeans==1,0], X_scaled[y_kmeans==1,1], label="Cluster 2")
plt.scatter(X_scaled[y_kmeans==2,0], X_scaled[y_kmeans==2,1], label="Cluster 3")
plt.scatter(X_scaled[y_kmeans==3,0], X_scaled[y_kmeans==3,1], label="Cluster 4")
plt.scatter(X_scaled[y_kmeans==4,0], X_scaled[y_kmeans==4,1], label="Cluster 5")

plt.title("Customer Segmentation using K-Means")
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.legend()

plt.show()
```

✓ ...



Step 9

Generate a dendrogram using hierarchical clustering to observe the hierarchical cluster formation.

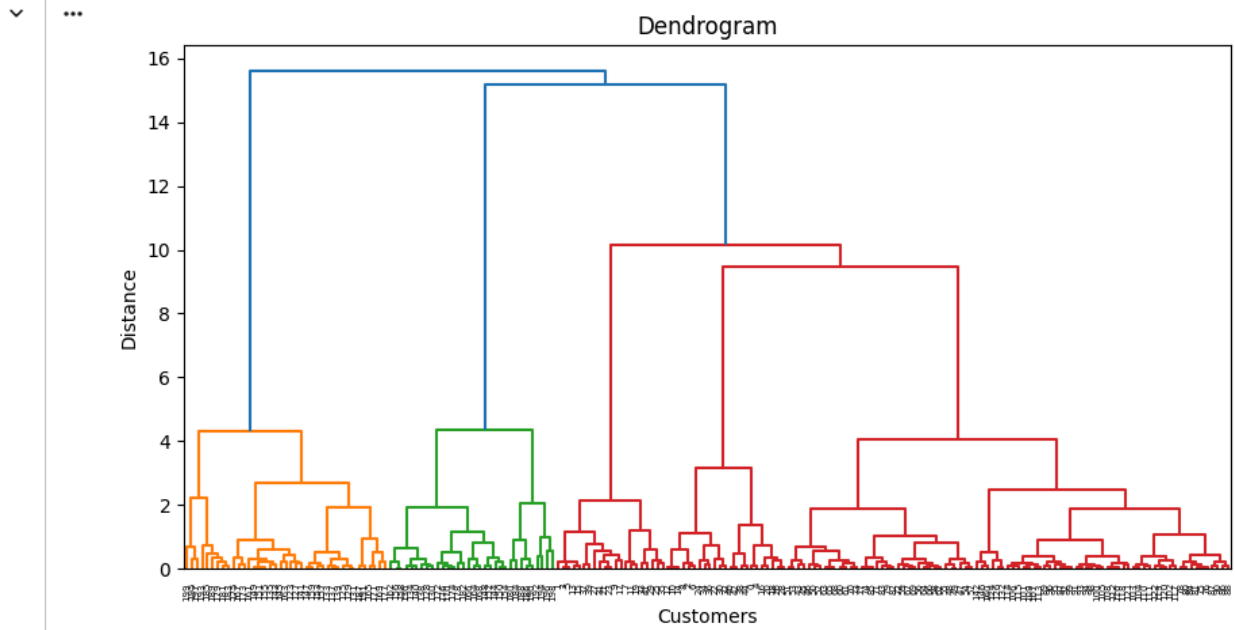
[10]
✓ 1s

```
plt.figure(figsize=(10,5))

dendrogram(linkage(X_scaled, method='ward'))

plt.title("Dendrogram")
plt.xlabel("Customers")
plt.ylabel("Distance")

plt.show()
```



Step 10

Apply Agglomerative Hierarchical Clustering and visualize the resulting clusters.

[11]
✓ Os

```
hc = AgglomerativeClustering(n_clusters=5, linkage='ward')

y_hc = hc.fit_predict(X_scaled)
```

[12]
✓ Os

```
plt.figure(figsize=(8,6))

plt.scatter(X_scaled[y_hc==0,0], X_scaled[y_hc==0,1], label="Cluster 1")
plt.scatter(X_scaled[y_hc==1,0], X_scaled[y_hc==1,1], label="Cluster 2")
plt.scatter(X_scaled[y_hc==2,0], X_scaled[y_hc==2,1], label="Cluster 3")
plt.scatter(X_scaled[y_hc==3,0], X_scaled[y_hc==3,1], label="Cluster 4")
plt.scatter(X_scaled[y_hc==4,0], X_scaled[y_hc==4,1], label="Cluster 5")

plt.title("Customer Segmentation using Hierarchical Clustering")
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.legend()

plt.show()
```



Performance Analysis

The Elbow Method indicated that **5 clusters** provide an optimal segmentation for the dataset.

K-Means clustering grouped customers into meaningful segments based on their income and spending patterns.

Typical clusters observed include:

High income – high spending customers

High income – low spending customers

Low income – high spending customers

Low income – low spending customers

Moderate income – moderate spending customers

Hierarchical clustering produced a similar grouping pattern and confirmed the presence of natural clusters in the dataset.

These clusters can help businesses identify target customer groups and develop effective marketing strategies.

Hyperparameter Tuning

For K-Means clustering, the main hyperparameter is the number of clusters K.

The Elbow Method was used to determine the optimal value of K by analyzing the WCSS plot.

Selected parameters:

```
n_clusters = 5  
init = k-means++  
max_iter = 300  
random_state = 42
```

For Hierarchical Clustering, the key parameters include:

```
n_clusters = 5  
linkage = ward
```

Ward linkage was selected because it minimizes intra-cluster variance and produces compact clusters.

Conclusion

In this experiment, K-Means and Hierarchical Clustering algorithms were applied to the Mall Customer Segmentation dataset to group customers based on their income and spending behavior.

The experiment demonstrated that clustering techniques can effectively identify customer segments, which can help businesses improve marketing strategies, customer targeting, and decision-making processes.